Syllabus

BIOL 4860/5860 — Experimental Design and Analysis

University of Massachusetts Lowell · 3 Credits · Spring 2027 Monday & Wednesday, 2:00–3:15 PM, Olsen Hall TBD

Instructor: Dr. Matthew Sasaki · **Office:** Olsen 205 · **Office Hours:** TBD, or by appointment · **Email:** Matthew_Sasaki@uml.edu

Prerequisites: Experimental Methods in Biology (BIOL.2330L) or graduate standing

Course description

Experiments are one of the main ways in which we expand our understanding of biology. In this hands-on course, students will develop fundamental skills in experimental design, data analysis, and visualization. Students will also learn how to use the coding language R to analyze and visualize data. By the end of the course, students will be able to describe experimental designs and analytic techniques (t-tests, ANOVA, linear regression, mixed effects models, etc.) commonly used in biological research. Students will also be able to evaluate the appropriateness of different experimental designs in biological research, design experiments to answer specific questions, and work through analytic workflows of medium complexity.

The course runs in three modules — see the full Schedule for week-by-week topics:

1. **R Crash Course** (Weeks 1–3) — introduction to R, data management best practices, and the typical analytic workflow for experimental work in the life sciences.
2. **Data Exploration** (Weeks 4–6) — working with data in R, focused on summarizing and visualizing key characteristics of biological data.
3. **Statistical Analysis and Interpretation** (Weeks 7–14) — commonly used experimental designs and the statistical techniques typically used to analyze biological data, pairing fundamental principles with applied practice each week. After Spring Break (Weeks 9–14) the module is organized around the idea that *common statistical tests are linear models*: t-tests, ANOVA, ANCOVA, correlation, chi-square tests, and (with extensions) mixed-effects models are all special cases of one framework. We interpret every model with the `marginaleffects` R package, which turns fitted models into predictions, comparisons, and slopes in the units of the data.

Learning objectives

At the end of this course, students will be able to:

- Apply principles of data management and project organization to facilitate the analysis of experimental data.
- Use the R coding language to import, clean, analyze, and visualize experimental data.
- Produce robust exploratory data analyses and visualizations for commonly encountered experimental designs.
- Discuss the appropriateness of different statistical tests for various experimental designs.

- Explain how common statistical tests (t-tests, ANOVA, correlation, chi-square) are special cases of the linear model, and fit them as such in R.
- Extend linear models to multiple predictors, interactions, non-normal responses (GLMs), and non-independent data (mixed-effects models).
- Interpret model results as estimates with uncertainty using marginal effects (predictions, comparisons, and slopes) rather than p-values alone.
- Interpret the results of key statistical analyses.

Required course materials

The primary requirement is access to **Canvas** and a **UML email account**. This course will also require *R for Data Science*, version 2, by Wickham, Çetinkaya-Rundel, and Grolemund. You're welcome to purchase a hard copy, but the text is also freely available online. We will also use free R packages including the `tidyverse`, `marginaleffects`, and `lme4` (installation instructions are in the first tutorials); the `marginaleffects` book is freely available online and Lindeløv's "Common statistical tests are linear models" is a recommended supplement. Personal computers are welcome, and access to the computer lab will also be provided.

Communication

All communication related to this course will come from the instructor's UML email or via Canvas announcements — please check both regularly. **Emails sent to the instructor must come from your UML email account.** For emails sent during the week, expect a response within 24 hours; if it's been more than 48 hours, a gentle reminder is welcome. Responses may be slower for emails sent over the weekend (Friday evening – Sunday evening) — please wait until Tuesday to follow up in that case.

Attendance, engagement, and classroom etiquette

Attendance — Lectures begin promptly at the start of the scheduled class period; arrive on time and stay for the full period. Regular attendance is one of the most important contributors to student success. If you need to miss class, contact Dr. Sasaki as soon as possible to discuss how to stay on track.

Illness — Please don't attend class if you're ill. Communicate your absence as soon as possible. You'll have an opportunity to make up missed assignments, but it's your responsibility to keep pace with lecture material.

Engagement and participation — Expect in-class questions, group discussions, and activities meant to build critical thinking and understanding. Reaching the correct answer matters, but the *process* of getting there matters more. All students are expected to be respectful and engaged — asking questions, taking good notes, giving full effort in group work, and paying attention when others are sharing.

Cell phones and laptops — Fine for notes or in-class activities; please avoid non-course use or distracting others. Persistent misuse may result in being asked to leave. Recording lectures without prior consent or a registered accommodation is not allowed.

Academic integrity policy

All students are advised of the university policy on academic integrity. Academic honesty is essential to maintaining trust in the learning environment and ensuring all students are evaluated fairly. Students are responsible for the honest completion and representation of their own work; academic dishonesty in

any form (plagiarism, cheating, fabrication, facilitating dishonesty) will not be tolerated and may result in disciplinary action, up to and including failure of the assignment or course, and referral to the provost's office.

AI statement

You are strongly discouraged from the use of AI and LLM agents (e.g., ChatGPT, Claude, etc.) in this course. While some uses — like proofreading written assignments or brainstorming ideas — can be beneficial, an increasingly common observation is that use of AI tools when learning new techniques (and especially when learning to code) blocks the development of skills and understanding.

Any AI usage must be documented in detail, and students must be able to describe how generated output was modified to reflect their own thoughts and perspective. Students are encouraged to think about (1) the consequences of AI usage for their own learning and for the environment, and (2) how AI use affects the development of critical thinking skills. As a general rule, **assignments generated entirely or primarily by AI will not be accepted**. In the event of suspected AI-generated responses, Dr. Sasaki may choose to check student understanding via a conversation and discuss how the answer was arrived at. After an initial warning, however, **violations of this policy will be considered academic misconduct**, subject to the university's Academic Integrity procedures.

Assignments

Over the course of the semester, you will complete a range of Assignments meant to develop your understanding of key points from the lecture materials. These assignments fall into three general categories:

Problem Sets – Students will complete weekly coding problem sets, questions about concepts and theory with application in both hypothetical and real-world scenarios. When analyzing example data sets, students will be asked to describe the methods and interpret results, as might be required when preparing scientific publications. Problem sets are graded on completion.

Data Safari Exercises – In these exercises, students are tasked to replicate a figure from the published literature using the publicly available data from that study. Students will also be asked to reflect on the process for each exercise: How easy was it to follow the published methods? How well does your plot match the published version? How would you change the paper, plot, or data to improve reproducibility?

Projects – In addition to the above components, *graduate students* are required to complete a project over the course of the semester. These projects will be broken down into the main analytic steps we will cover in this course – experimental design and planned hypothesis testing, data carpentry and exploration, statistical modelling, and visualization. Ideally, students will complete this project using their own data, but data obtained from the published literature is also acceptable. Projects will be presented at the end of the semester.

Assignments are distributed as **Quarto** (`.qmd`) templates, downloadable from the Schedule. Complete them in the template, then render to **PDF** (via Typst, bundled with Quarto) and submit both the `.qmd` and the `.pdf`. Given the pacing and breadth of this course, it is important to turn work in on time. Late assignments can be turned in for credit up to three weeks following the deadline, but will be **docked 2% for every day past the due date (weekends included)**.

Exams and quizzes

There are no midterm exams for this course. Each unit includes several short, in-class, open-note quizzes to gauge understanding of key concepts.

Attendance & participation

Both regular attendance and active participation are essential given the nature of this course. Attendance is recorded using a combination of in-class check-ins/reflection assignments or data collection exercises. Participation points come from activities including (but not limited to) in-class discussions, out-of-class surveys and data collection exercises, and office hour visits.

Grading

Undergraduates: Assignments 50%, Quizzes 40%, Attendance & Participation 10%.

Graduate students: Project 50%, Assignments 25%, Quizzes 20%, Attendance & Participation 5%.

Undergraduate scale

Grade	Percentage
A	94–100
A-	90–93
B+	87–89
B	83–86
B-	80–82
C+	77–79
C	73–76
C-	70–72
D	60–69
F	< 60

Graduate scale

Grade	Percentage
A	94–100
A-	90–93
B+	87–89
B	83–86
B-	80–82
C+	76–79
F	< 75

Student mental health and well-being

Your personal health and well-being are important to all of us at the university.

Service / Office	Description	Contact
Counseling Services	Crisis intervention, assessment, referrals, short-term individual counseling and group therapy	(978) 934-6800
UMatter2	University-wide initiative supporting mental health, including the Togetherall peer-to-peer platform	(978) 934-6671
CLASS Advising	Advising services: goal setting, course selection, SIS functions, major/minor changes, course deletions	(978) 934-2936 · Advisement@uml.edu
Office of Student Life & Wellbeing	Advances holistic student success through health-promoting actions and campus culture	(978) 934-4342 · Wellbeing@uml.edu

Disability services

If you are registered with Disability Services and require course accommodations, notify Dr. Sasaki via the Accommodate request process as soon as possible so appropriate arrangements can be made. Please connect during office hours or privately after class to discuss logistics. For more information or to register, visit the Disability Services website.

University privacy statement

UMass Lowell recognizes the importance of mutual trust between students and faculty. Massachusetts is a two-party consent state, meaning it is illegal to record someone without their permission. Recordings of classroom lectures are the instructor's intellectual property; instructors may prohibit audio/video recording of their lectures unless the requesting student is registered with Disability Services and recording is an approved accommodation. Sharing or selling recordings of classroom activities, discussions, or lectures without the instructor's permission is prohibited and may result in sanctions for academic dishonesty or referral to the dean's or provost's office.

Course Schedule

Course schedule subject to change — see the Schedule page for the current, up-to-date version.